Golden Coordinates at Level Five: Ruled-Surface Rank Loss, CM Values, and Fourier Self-Duality

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20 October 2026
Local assembled draft

Abstract

The rank-loss polynomial of an orthogonal-circle ruled surface is c^2-3c+1, with interior root $c_0=(3-\sqrt{5})/2$. We ask for actual maps linking this golden coordinate to level-five modular and Fourier phenomena. A certified Klein-equation factorization at $\tau_{163}=(1+\mathrm{i}\sqrt{163})/2$ recovers $\sqrt{5}$ from $v=r(\tau_{163})^5$. Duke's inversion at the distinct point i becomes the normalized sine-Fourier reflection $A=JMJ$, and Poisson–Mellin analysis carries its positive eigenline to a functional equation and classifies the two signed functional equations within the odd five-periodic coefficient space. Exact finite certificates are kept distinct from the analytic proofs.

1 Question and surface coordinate

Hermite's 1859 discussion leads, in modern notation, to the remarkably small gap between $\mathrm{e}^{\pi\sqrt{163}}$ and an integer [4, pp. 47–48]. For $\tau_{163}=(1+\mathrm{i}\sqrt{163})/2$ and $q=\mathrm{e}^{2\pi\mathrm{i}\tau_{163}}=-\mathrm{e}^{-\pi\sqrt{163}}$, the classical j-expansion $q^{-1}+744+196884q+\cdots$ and $j(\tau_{163})=-640320^3$ explain its small gap [3]. Meanwhile $P(c)=c^2-3c+1$ has discriminant 5, so its roots lie in $\mathbb{Q}(\sqrt{5})=\mathbb{Q}(\zeta_5+\zeta_5^{-1})$. Indeed $\zeta_5+\zeta_5^{-1}=2\cos(2\pi/5)=(\sqrt{5}-1)/2$ has minimal polynomial X^2+X-1, identifying the real quadratic cyclotomic subfield. Which calculations connect these facts?

Put $\phi=(1+\sqrt{5})/2$, $\gamma=\phi^{-1}$, $c_0=\phi^{-2}=\gamma^2$. The local surface source [5] uses the surface angle t and ruling coordinate u:

$$S(t,u)=(c+2u(1-c),(1-u)\sin t,u\sqrt{c(2-c)}),\qquad c=\cos t. \tag{1}$$

The paired endpoints are $C_1(t)=(c,\sin t,0)$ and $C_2(t)=(2-c,0,\sqrt{c(2-c)})$: unit circles in perpendicular planes, with centers $(0,0,0)$ and $(1,0,0)$. Thus $S(t,u)=(1-u)C_1(t)+uC_2(t)$ for $0\le u\le 1$. For fixed t, varying u traces a ruling segment. Rank loss means that its two parameter-direction tangent vectors become linearly dependent. In $0<c<1$, the third component of $S_t\times S_u$ is $(1-c)(1-c-2u)$, hence rank loss forces $u=(1-c)/2$. The remaining components are

$$-\frac{(1+c)P(c)}{2\sqrt{c(2-c)}},\qquad -\frac{\sin t\,P(c)}{\sqrt{c(2-c)}}.$$

Thus

$$c_0=\frac{3-\sqrt{5}}{2},\qquad \gamma=1-c_0=\phi^{-1},\qquad u_0=\gamma/2=\cos(2\pi/5). \tag{2}$$

Figures 1–5 display the connection map, the ruled patch and its rank-loss locus, and the pentagonal and icosahedral realizations.

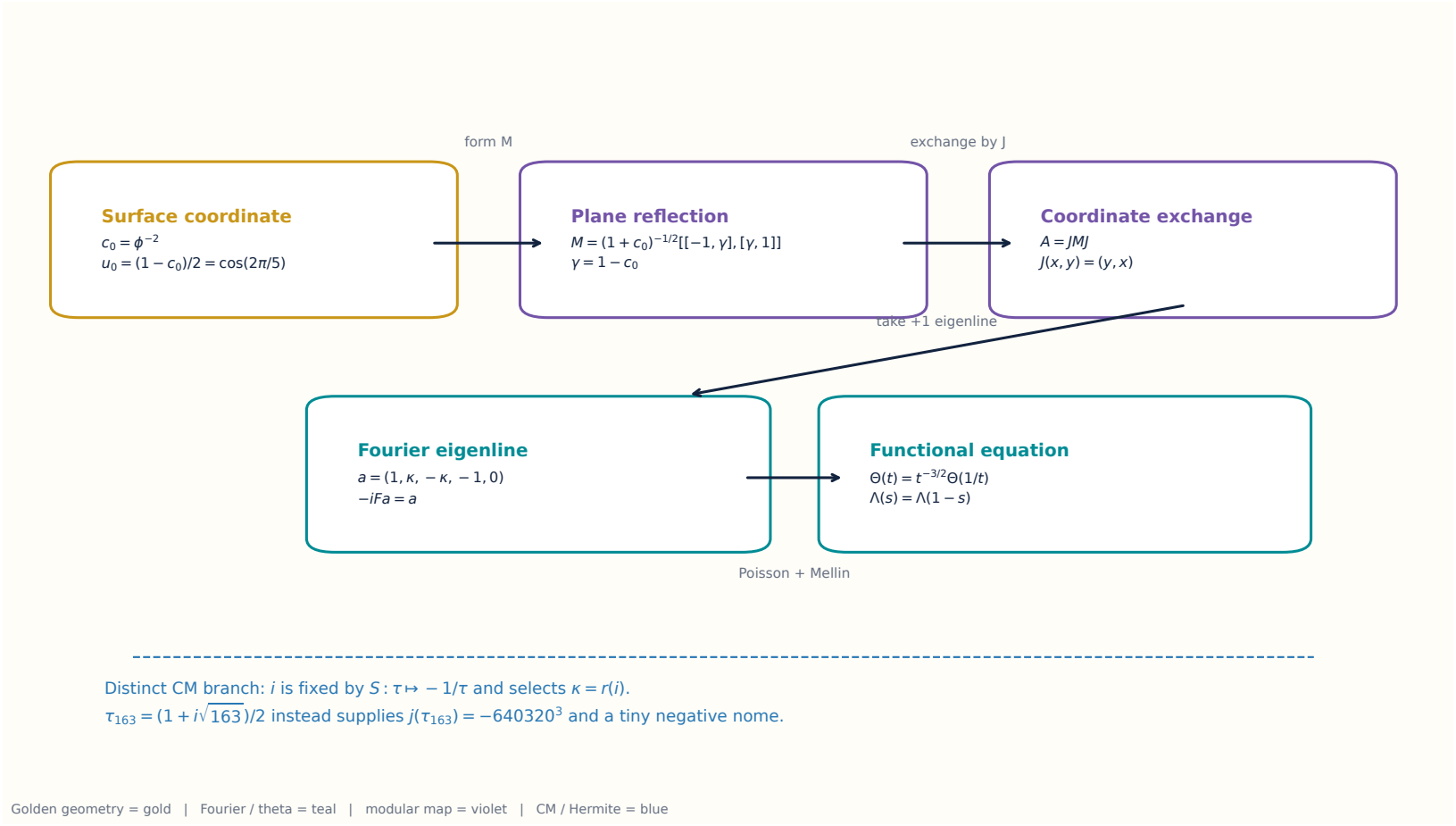


Figure 1: Connection map: surface coordinate, reflection, coordinate exchange, Fourier eigenline, and completed functional equation.

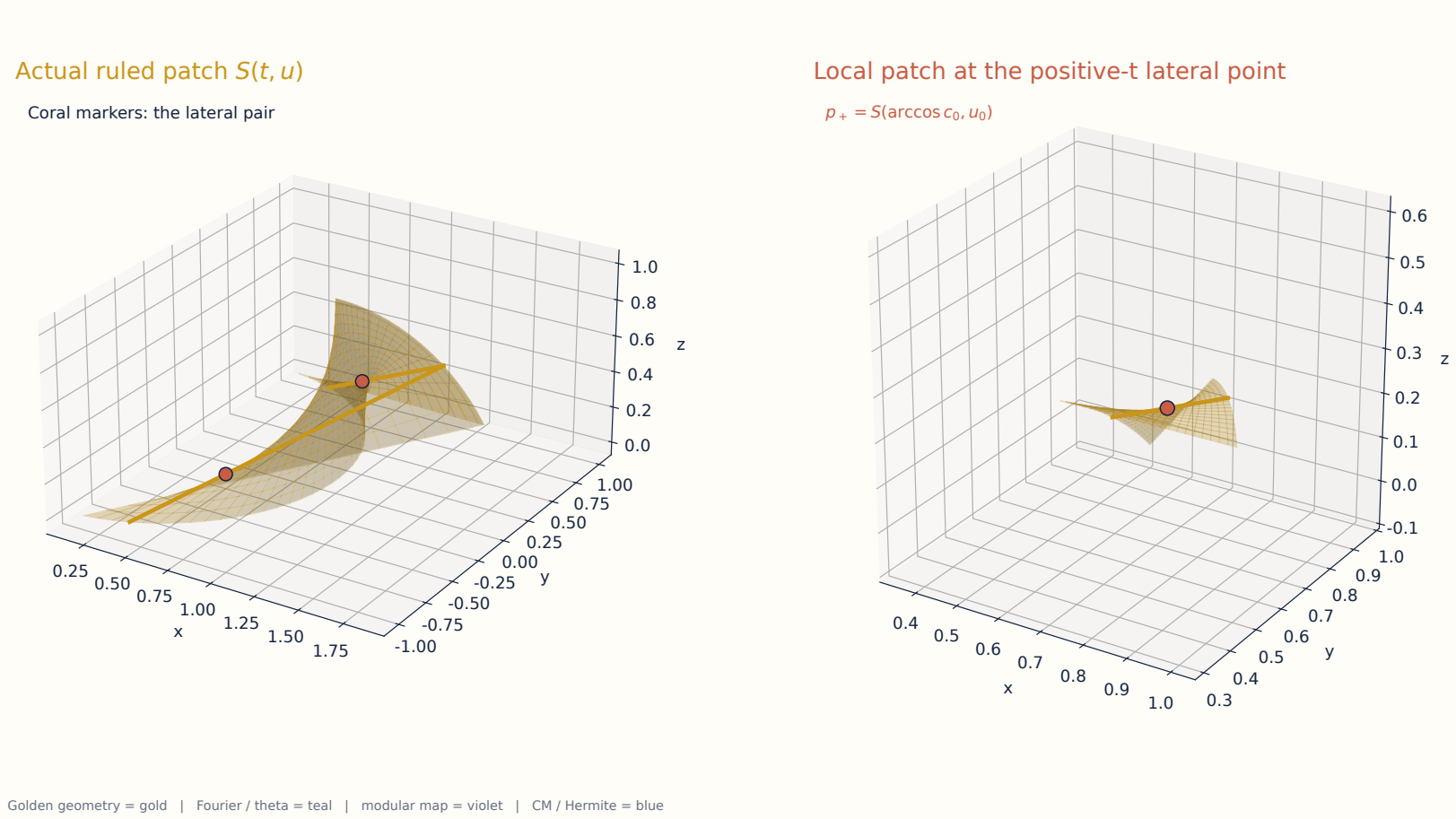


Figure 2: Orthogonal-circle ruled surface (1): a global reflected lateral pair at $t = \pm \arccos c_0$ and a local positive-t close-up at the rank-loss ruling u_0.

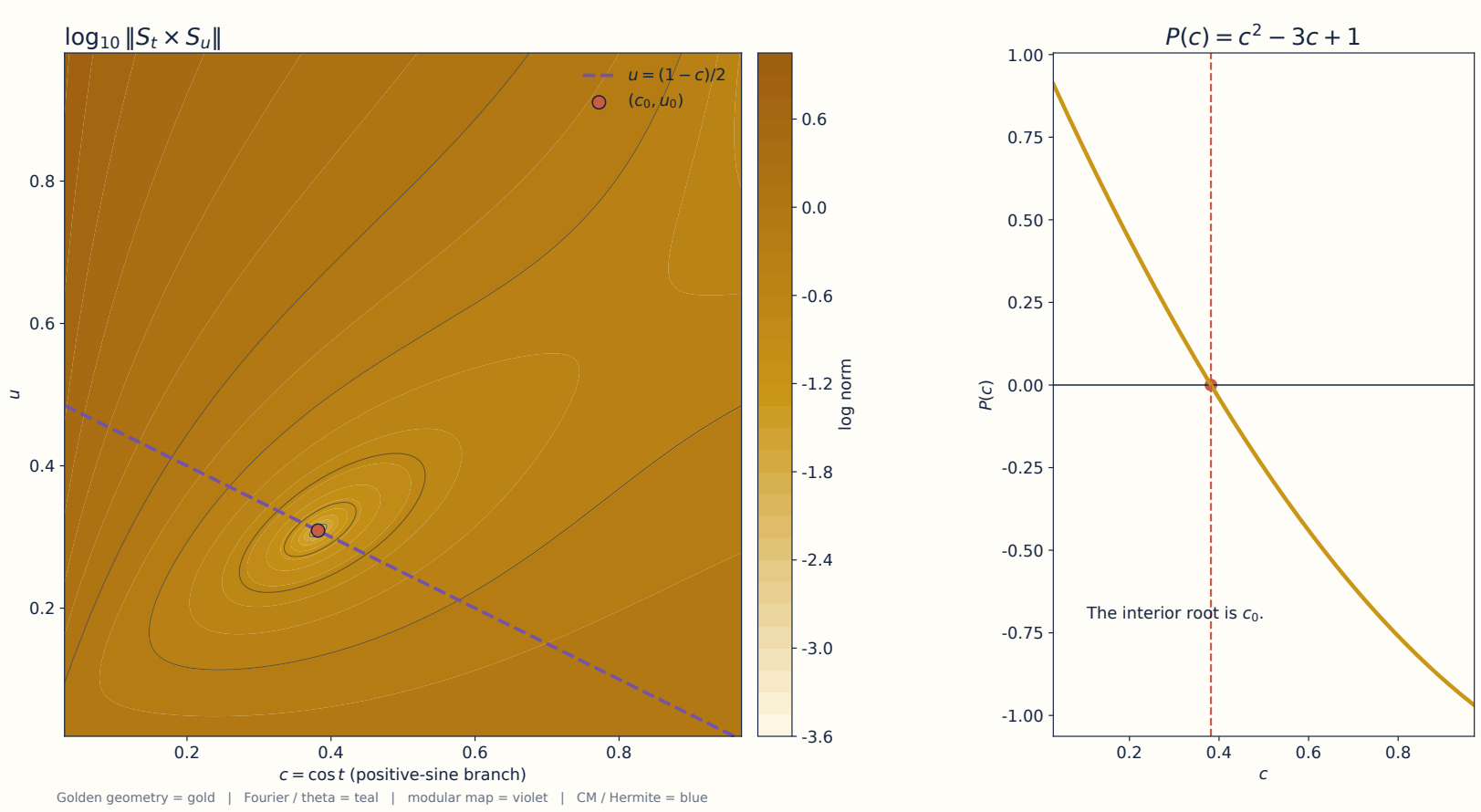


Figure 3: Rank loss in (c, u): the positive-sine branch, $u = (1 - c)/2$, the remaining polynomial $P(c)$, and its root c_0.

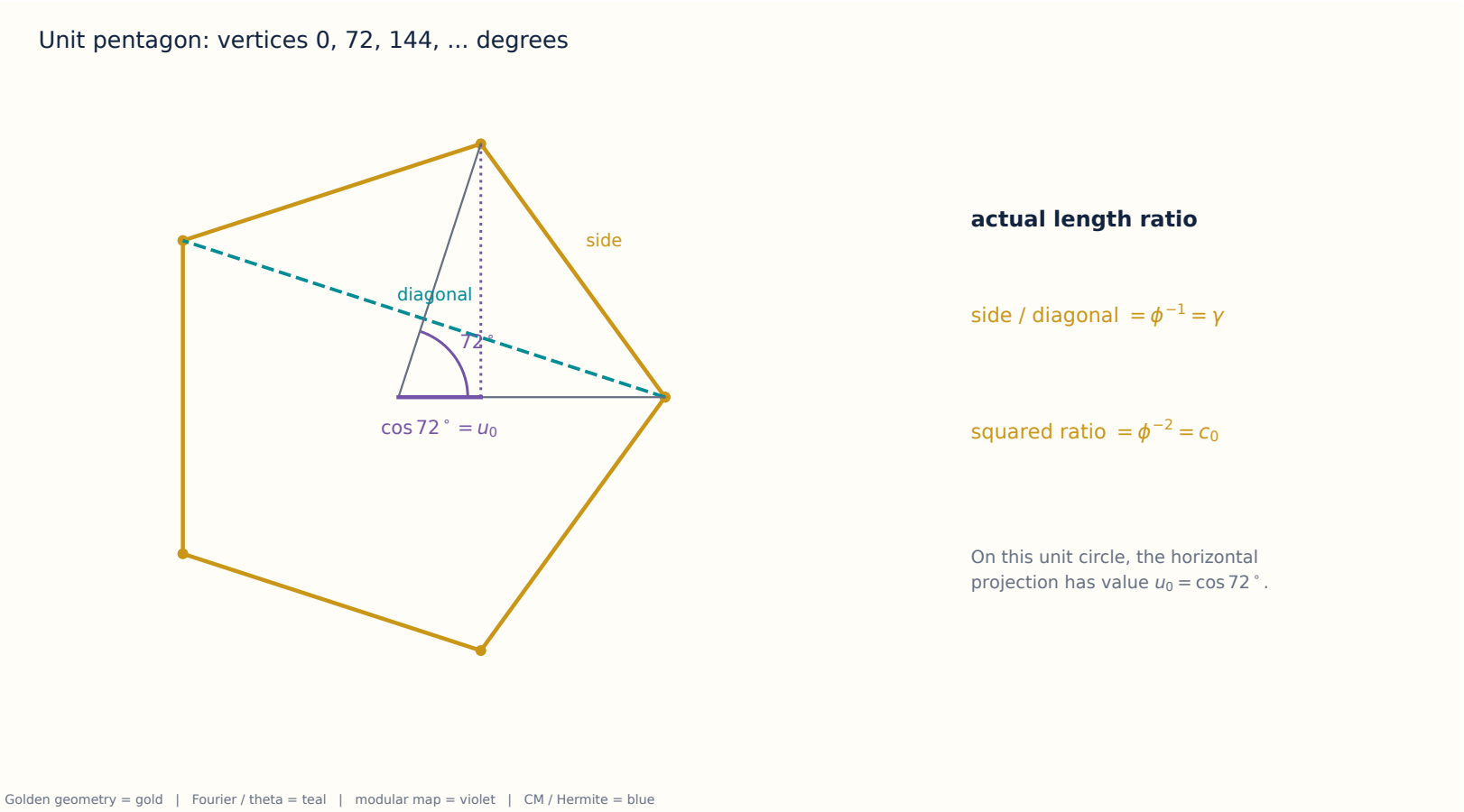


Figure 4: Unit-pentagon coordinate: $u_0 = \cos 72^\circ$ is kept distinct from c_0, the squared side-to-diagonal ratio.

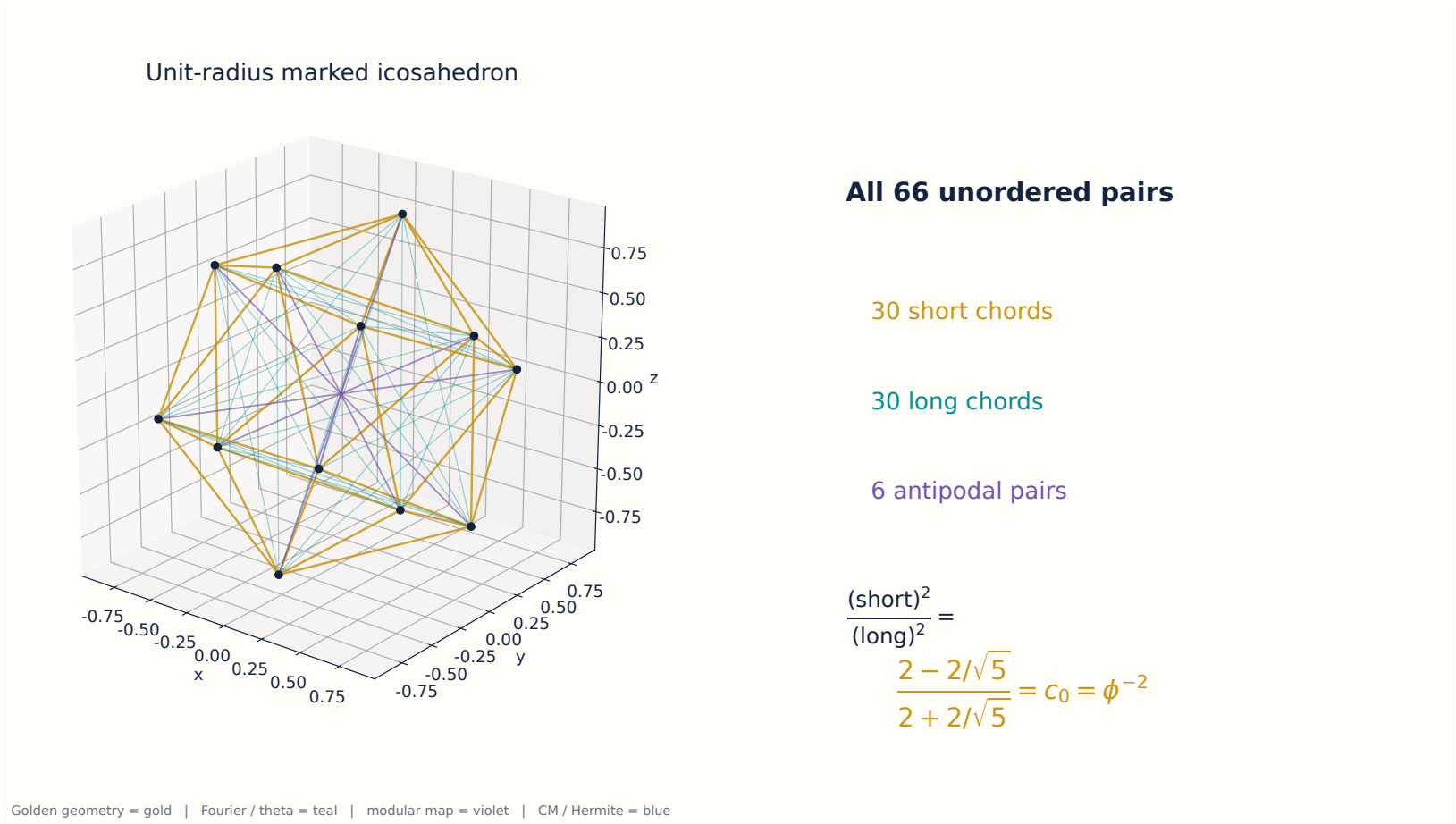


Figure 5: Unit-radius icosahedron chords. The vertices are $\{(0,\pm1,\pm\phi),(\pm1,\pm\phi,0),(\pm\phi,0,\pm1)\}/\sqrt{1+\phi^2}$. Their distinct dot products are $1/\sqrt{5}, -1/\sqrt{5}, -1$, with pair counts $30, 30, 6$. Thus the squared short/long ratio is $(2-2/\sqrt{5})/(2+2/\sqrt{5}) = c_0$.

2 CM field recovery

For $\Im\tau > 0$ define

$$r(\tau) = q^{1/5} \prod_{n\geq 1} \frac{(1-q^{5n-4})(1-q^{5n-1})}{(1-q^{5n-3})(1-q^{5n-2})}, \quad q = \mathrm{e}^{2\pi \mathrm{i}\tau}, \quad q^{1/5} = \mathrm{e}^{2\pi \mathrm{i}\tau/5}. \tag{3}$$

Set $v = r(\tau_{163})^5$. The classical CM value above and Klein's level-five relation (as recorded by Duke [3, Eq. (2.5)]) give

$$P_j(v) = (v^4 - 228v^3 + 494v^2 + 228v + 1)^3 + jv(v^2 + 11v - 1)^5 = 0. \tag{4}$$

Theorem 2.1. *For $v = r(\tau_{163})^5$, $\sqrt{5} \in \mathbb{Q}(v)$ and*

$$\sqrt{5} = -\frac{A_{163}(v)}{B_{163}(v)}, \qquad \phi = \frac{B_{163} - A_{163}}{2B_{163}},$$
$$\gamma = -\frac{A_{163} + B_{163}}{2B_{163}}, \quad c_0 = \frac{3B_{163} + A_{163}}{2B_{163}}.$$

where the last three expressions are evaluated at v.

Proof. Appendix A gives the exact identity $P_{-640320^3}(X) = A_{163}(X)^2 - 5B_{163}(X)^2$ with coprime integer polynomials. If $B_{163}(v) = 0$, then $A_{163}(v) = 0$, contrary to Bezout. Thus $(A_{163}(v)/B_{163}(v))^2 = 5$. The rational interval certificate gives $A_{163}(v) < 0 < B_{163}(v)$, selecting $-\sqrt{5}$; the formulas follow. □

Figure 6 compares the two CM arguments, nomes, and j-values; Figure 7 magnifies the near-integer gap and its series corrections.

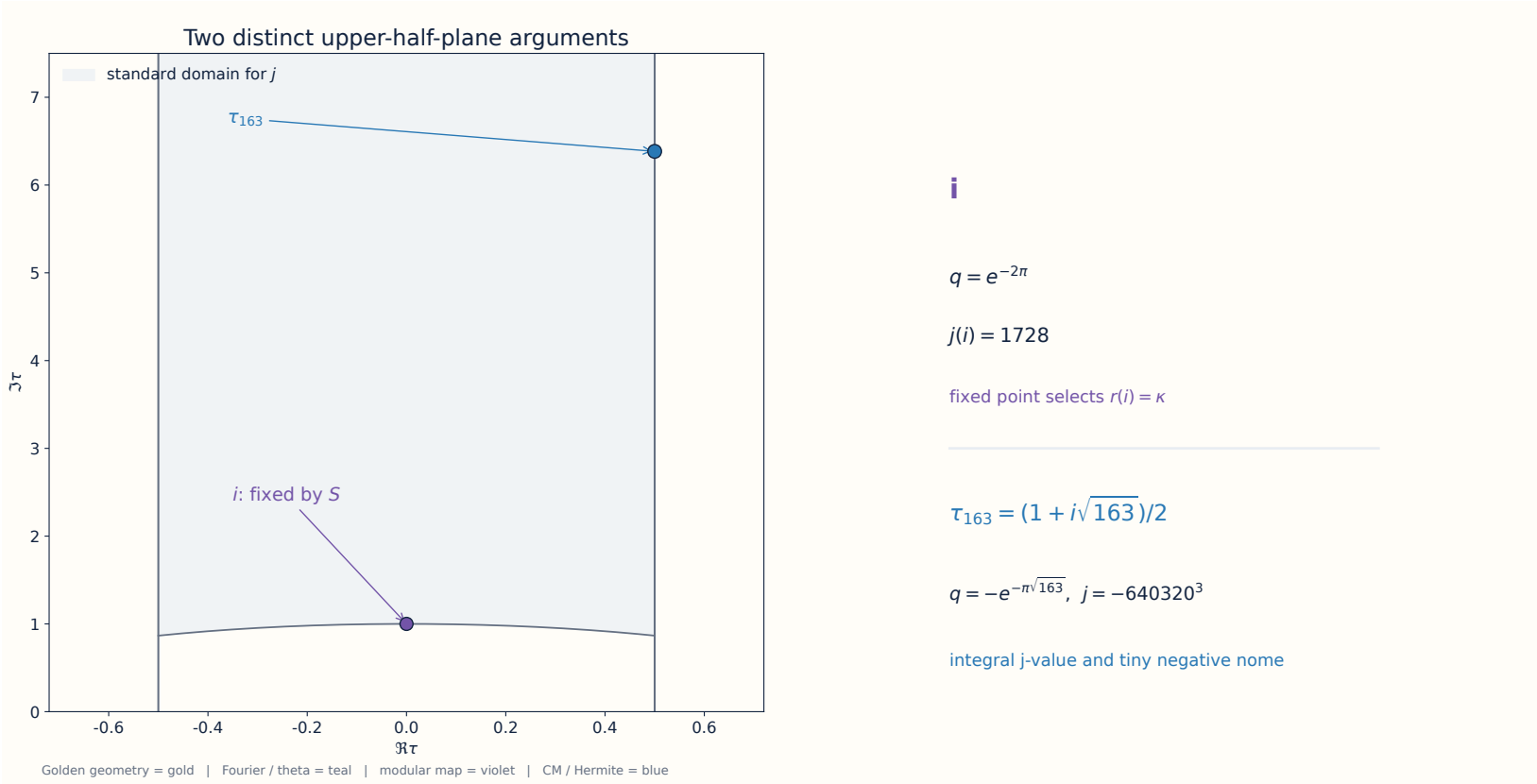


Figure 6: Two CM arguments, their nomes, and their j-values. The point i is the inversion fixed point; the tiny negative nome at τ_{163} is the arithmetic branch in Theorem 2.1.

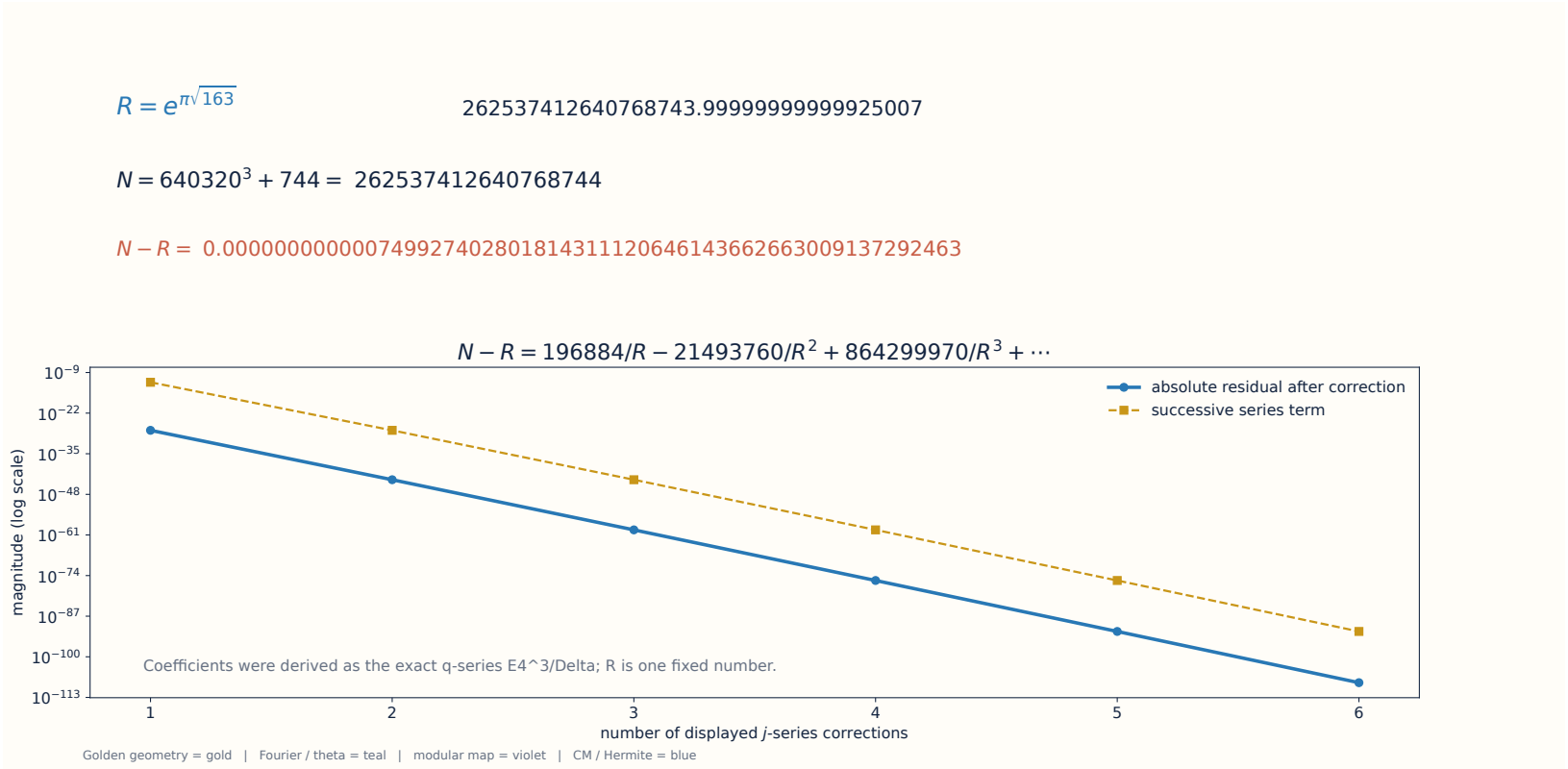


Figure 7: Hermite near-integer profile: the fixed gap and successive j-series corrections, separate from the exact field certificate.

3 The common involution

Duke's inversion law is

$$r(-1/\tau) = \frac{\gamma - r(\tau)}{1 + \gamma r(\tau)} [3, \textit{Proposition } 2].$$

In homogeneous coordinates put

$$M = \frac{1}{\sqrt{1+c_0}} \begin{pmatrix} -1 & \gamma \\ \gamma & 1 \end{pmatrix}, \qquad J = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Taking the ratio of $M(z,1)^T$ gives the displayed law, and direct multiplication, using $\gamma^2 = c_0$, proves $M^2 = I$.

Let V_5 be the real space of odd five-periodic sequences. In residues $1,2,3,4,5$ (with $5 = 0$), it consists of $(x, y, -y, -x, 0)$. Define

$$(\mathcal{F}a)_m = \frac{1}{\sqrt{5}} \sum_{n=1}^{5} a_n \mathrm{e}^{2\pi \mathrm{i} mn/5}, \qquad \mathcal{T} = -\mathrm{i}\mathcal{F}.$$

Pairing n and $5-n$ gives

$$A = \frac{2}{\sqrt{5}} \begin{pmatrix} \sin(2\pi/5) & \sin(\pi/5) \\ \sin(\pi/5) & -\sin(2\pi/5) \end{pmatrix} = JMJ. \tag{5}$$

Indeed $2\sin(2\pi/5)/\sqrt{5} = 1/\sqrt{1+c_0}$ and $2\sin(\pi/5)/\sqrt{5} = \gamma/\sqrt{1+c_0}$, by squared pentagon identities. Thus $A^2 = I$.

Proposition 3.1. *With $\kappa = r(i)$,*

$$\kappa = \frac{\gamma}{1+\sqrt{1+c_0}}, \qquad A(1,\kappa)^T = (1,\kappa)^T.$$

For $\theta = \frac{1}{2}\arctan\gamma$,

$$\kappa = \tan\theta, \quad \cos 2\theta = \frac{1}{\sqrt{1+c_0}}, \quad \sin 2\theta = \frac{\gamma}{\sqrt{1+c_0}},$$

and, separately, $\kappa = 2\cos 18^\circ - 2\cos 36^\circ$. Finally $u_0 = \kappa/(1-\kappa^2)$ and $c_0 = (2\kappa/(1-\kappa^2))^2$.

Proof. The inversion law fixes $r(i)$; positivity of every product factor in (3) at i selects the positive root of $\gamma X^2 + 2X - \gamma = 0$. It gives the stated κ and $M(\kappa,1)^T = (\kappa,1)^T$, so J transfers it explicitly to the $+1$ eigenvector of A. Since $\tan 2\theta = \gamma$ and $0 < 2\theta < \pi/2$, right-triangle and half-angle formulas prove the three half-angle identities. This axis angle is not 18°. Separately $2\cos 36^\circ = \phi$, $2\cos 18^\circ = \sqrt{2+\phi}$; squaring positive sides shows their difference has the same positive quadratic root. The last formulas are the tangent double-angle identity. □

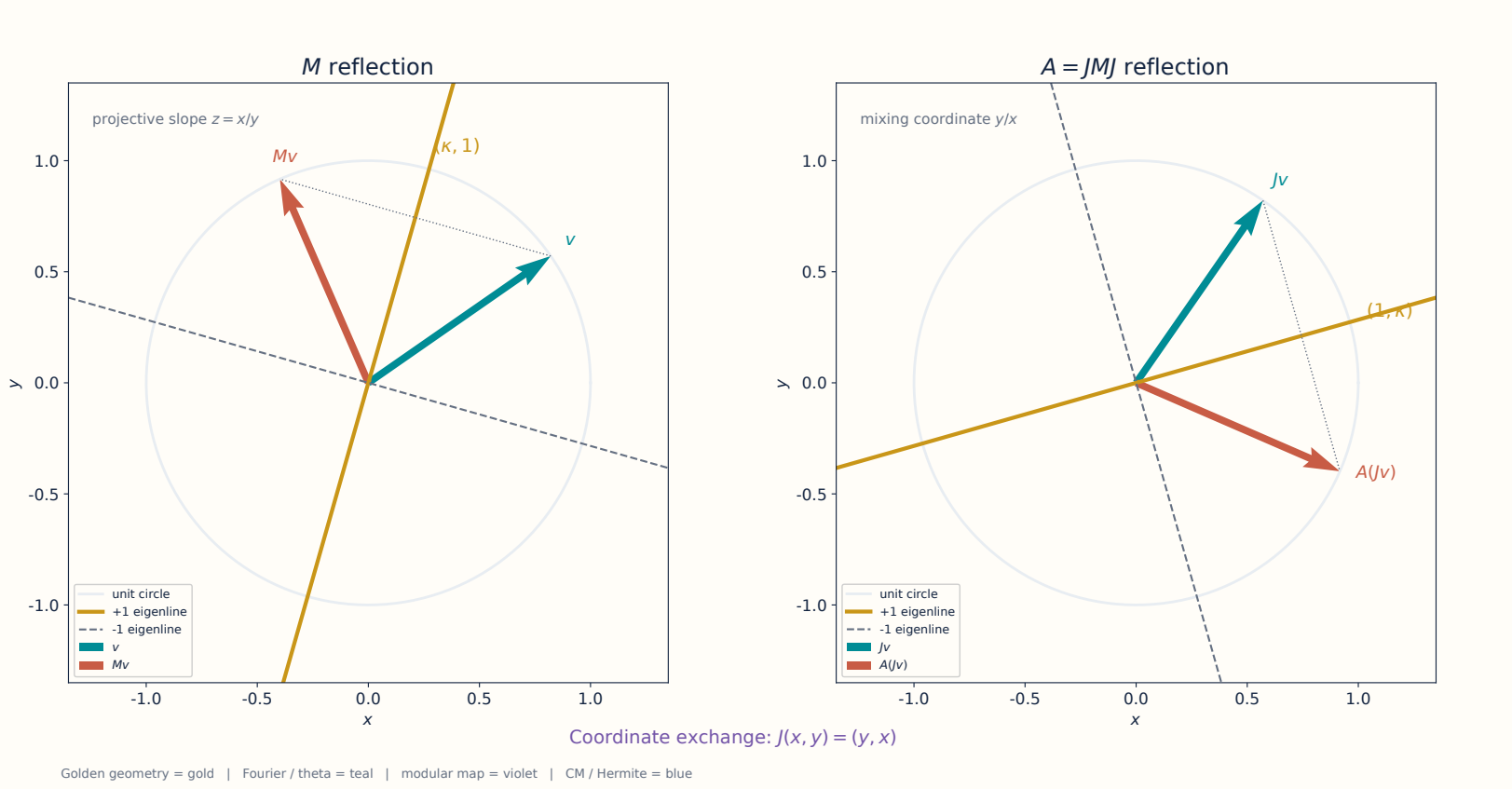


Figure 8: Coordinate exchange for M and $A = JMJ$, using an off-axis vector and keeping their slope conventions separate.

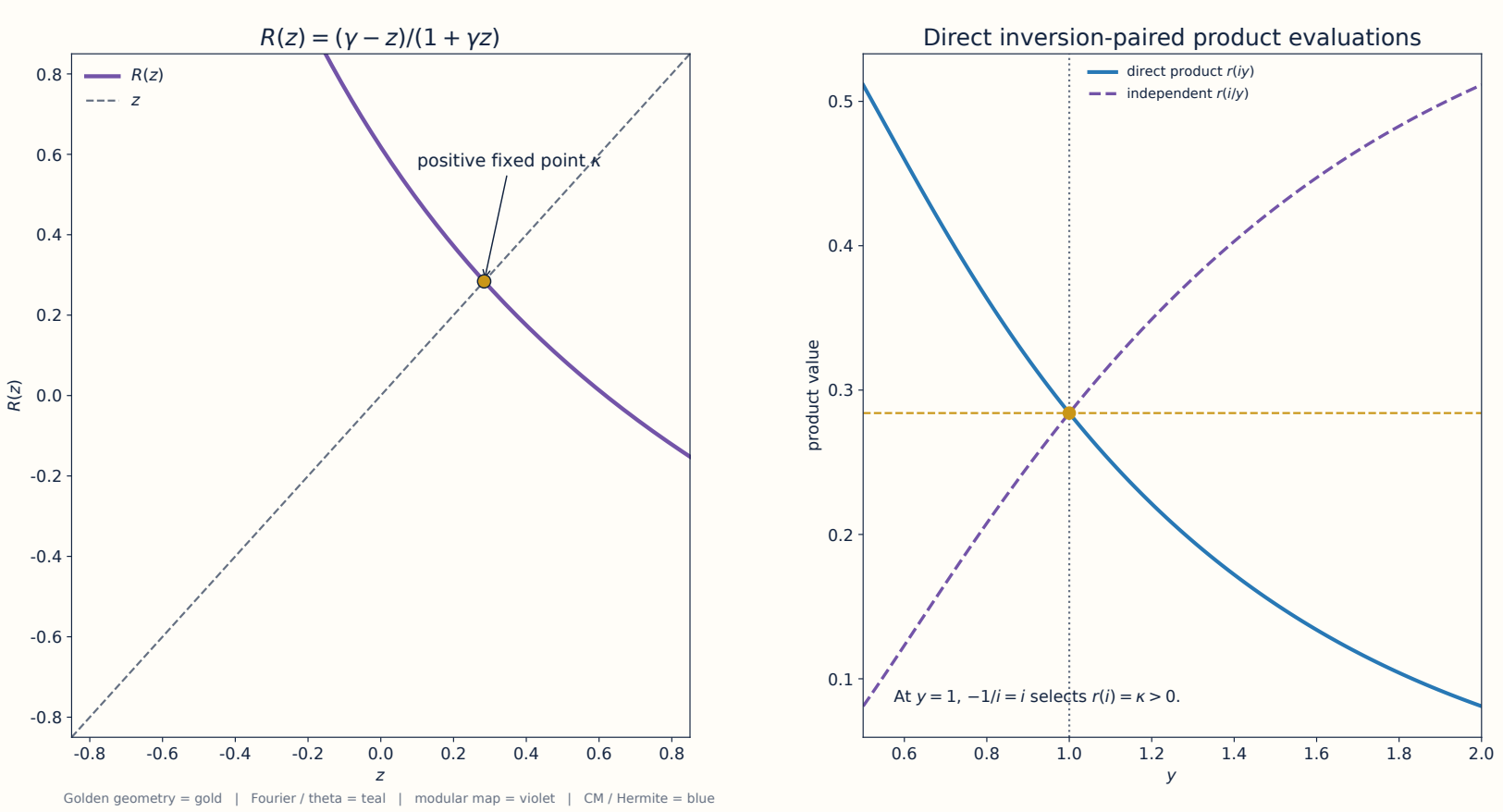


Figure 9: Rogers–Ramanujan inversion at the positive fixed point: independent direct product curves $r(\mathrm{i}y)$ and $r(\mathrm{i}/y)$ meet at $r(i) = \kappa$.

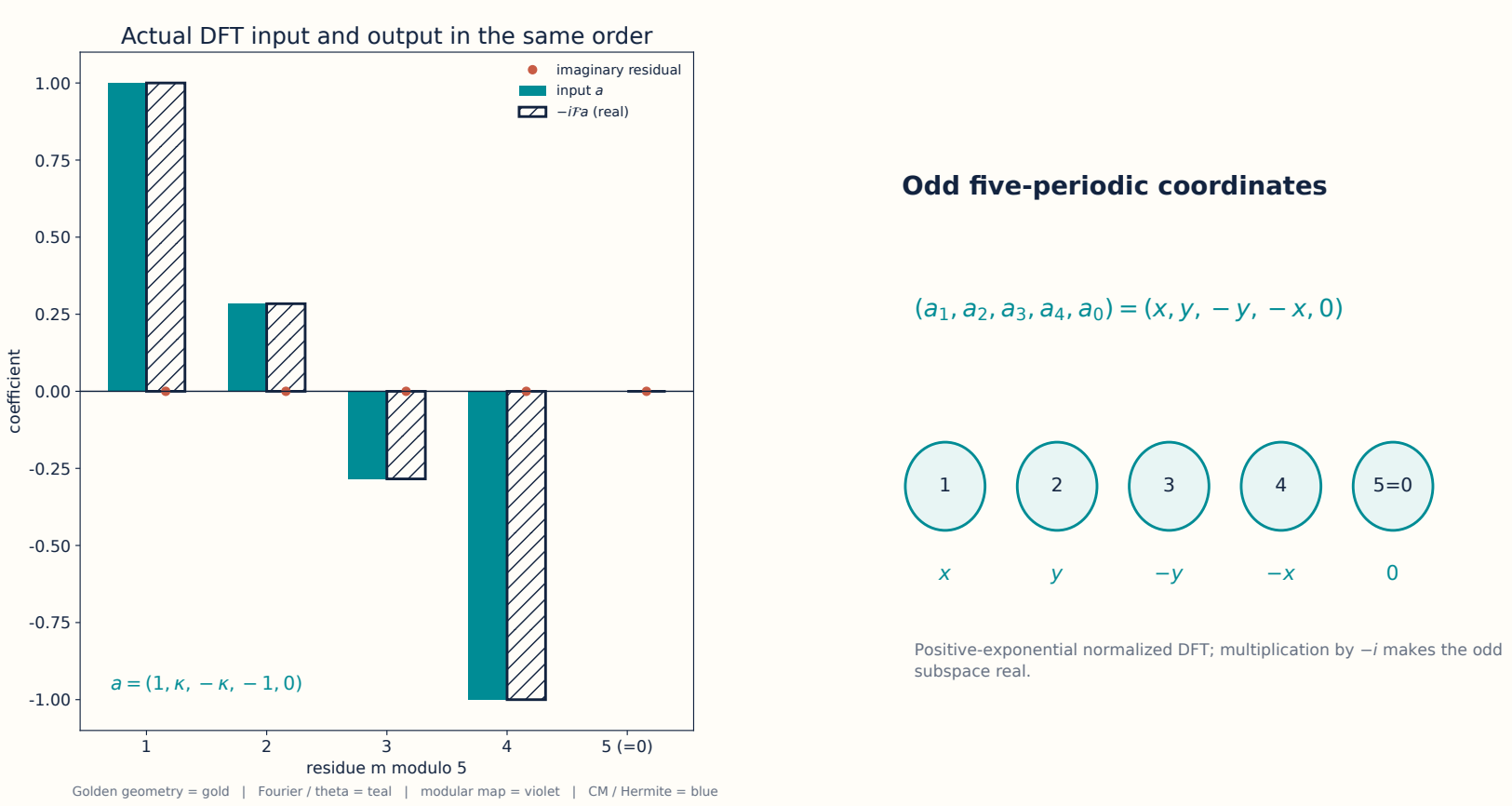


Figure 10: Odd five-periodic Fourier eigenvector in residue order $1, 2, 3, 4, 5$; the plotted input and output are the actual normalized DFT vectors.

Also

$$X^4 + 2X^3 - 6X^2 - 2X + 1 = (X^2 + 2\phi X - 1)(X^2 - 2\phi^{-1}X - 1),$$

the polynomial norm $N_{\mathbb{Q}(\sqrt{5})(X)/\mathbb{Q}(X)}(X^2+2\phi X-1)$. Conjugation sends ϕ to $-\phi^{-1}$; multiplication using $\phi-\phi^{-1}=1$ and $\phi\phi^{-1}=1$ proves the identity. The fixed-point equation gives $\kappa^2+2\phi\kappa-1=0$, so the first factor vanishes at $X=\kappa$.

4 Poisson–Mellin bridge and classification

The Gaussian bridge turns finite Fourier symmetry into analytic symmetry. To retain t as a surface angle, use $x>0$ here:

$$F_a(s) = \sum_{n\geq 1} a_n n^{-s} \quad (\Re s > 1), \qquad \Theta_a(x) = \sum_{n\in\mathbb{Z}} n a_n \mathrm{e}^{-\pi n^2 x/5}.$$

Theorem 4.1. *For $a \in V_5$, $a' = \mathcal{T}a$,*

$$\Lambda_a(s) = \frac{1}{2}\int_0^\infty \Theta_a(x) x^{(s+1)/2}\frac{dx}{x} = \left(\frac{5}{\pi}\right)^{(s+1)/2}\Gamma((s+1)/2)F_a(s)$$

defines an entire function by its integral; the displayed Dirichlet-series expression holds for $\Re s > 1$. Moreover $\Lambda_a(s) = \Lambda_{a'}(1-s)$.

Proof. With $\widehat{f}(\xi) = \int_{\mathbb{R}} f(y)\mathrm{e}^{-2\pi \mathrm{i} y\xi}dy$, put $f_x(y) = y\mathrm{e}^{-\pi x y^2/5}$. Its transform is

$$\widehat{f_x}(\xi) = -\mathrm{i}(5/x)^{3/2}\xi \mathrm{e}^{-5\pi\xi^2/x}.$$

Residue-class Poisson summation [2, §1.8(iv)] is

$$\sum_{\ell\in\mathbb{Z}} f_x(j+5\ell) = \frac{1}{5}\sum_{k\in\mathbb{Z}} \mathrm{e}^{2\pi \mathrm{i} kj/5}\widehat{f_x}(k/5).$$

After multiplying by a_j and summing,

$$\begin{aligned}\Theta_a(x) &= \frac{1}{\sqrt{5}}\sum_k (\mathcal{F}a)_k \widehat{f_x}(k/5) \\ &= \frac{1}{\sqrt{5}}\sum_k \mathrm{i}a'_k[-\mathrm{i}(5/x)^{3/2}(k/5)\mathrm{e}^{-\pi k^2/(5x)}] = x^{-3/2}\Theta_{a'}(1/x).\end{aligned}$$

This shows the $1/5$, $1/\sqrt{5}$, $\mathrm{i}(-\mathrm{i})$ cancellation, and $x^{-3/2}$. Bounded periodicity gives exponential decay at infinity; the transformation gives $Cx^{-3/2}\mathrm{e}^{-c/x}$ at zero. These envelopes, also after multiplication by $|\log x|^r$, prove local uniform convergence of the integral and all s derivatives. For $\Re s > 1$, odd pairing and termwise gamma integration give the displayed completion. Substitution of the theta transformation changes $x^{(s+1)/2}$ to $x^{(s-2)/2}$, and $y = 1/x$ gives $y^{(2-s)/2}dy/y$, proving the functional equation. □

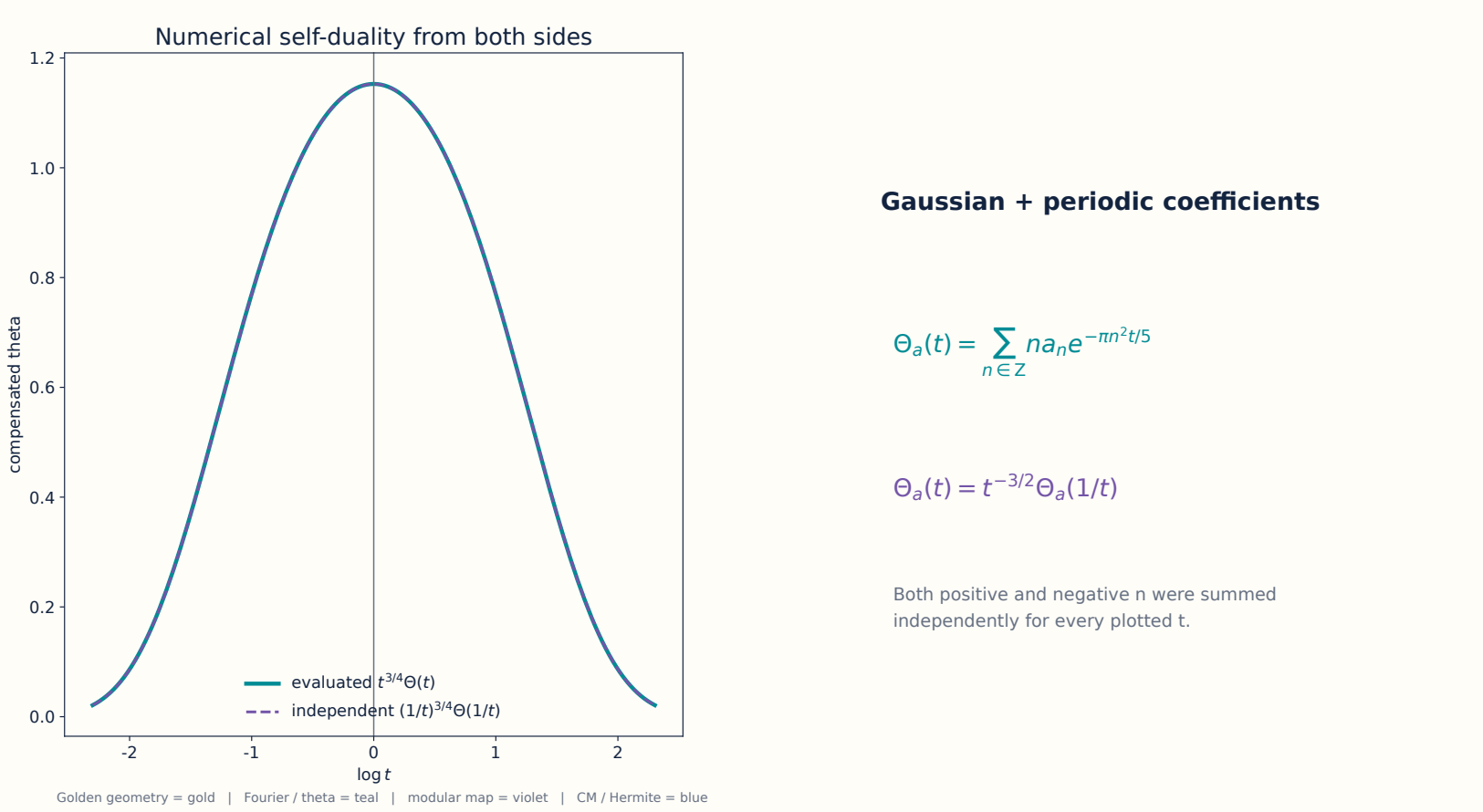


Figure 11: Compensated theta self-duality: the profile $x^{3/4}\Theta_a(x)$ is evaluated independently at reciprocal arguments and is even as a function of $\log x$ (the plot labels its positive scale by t), as predicted by Theorem 4.1.

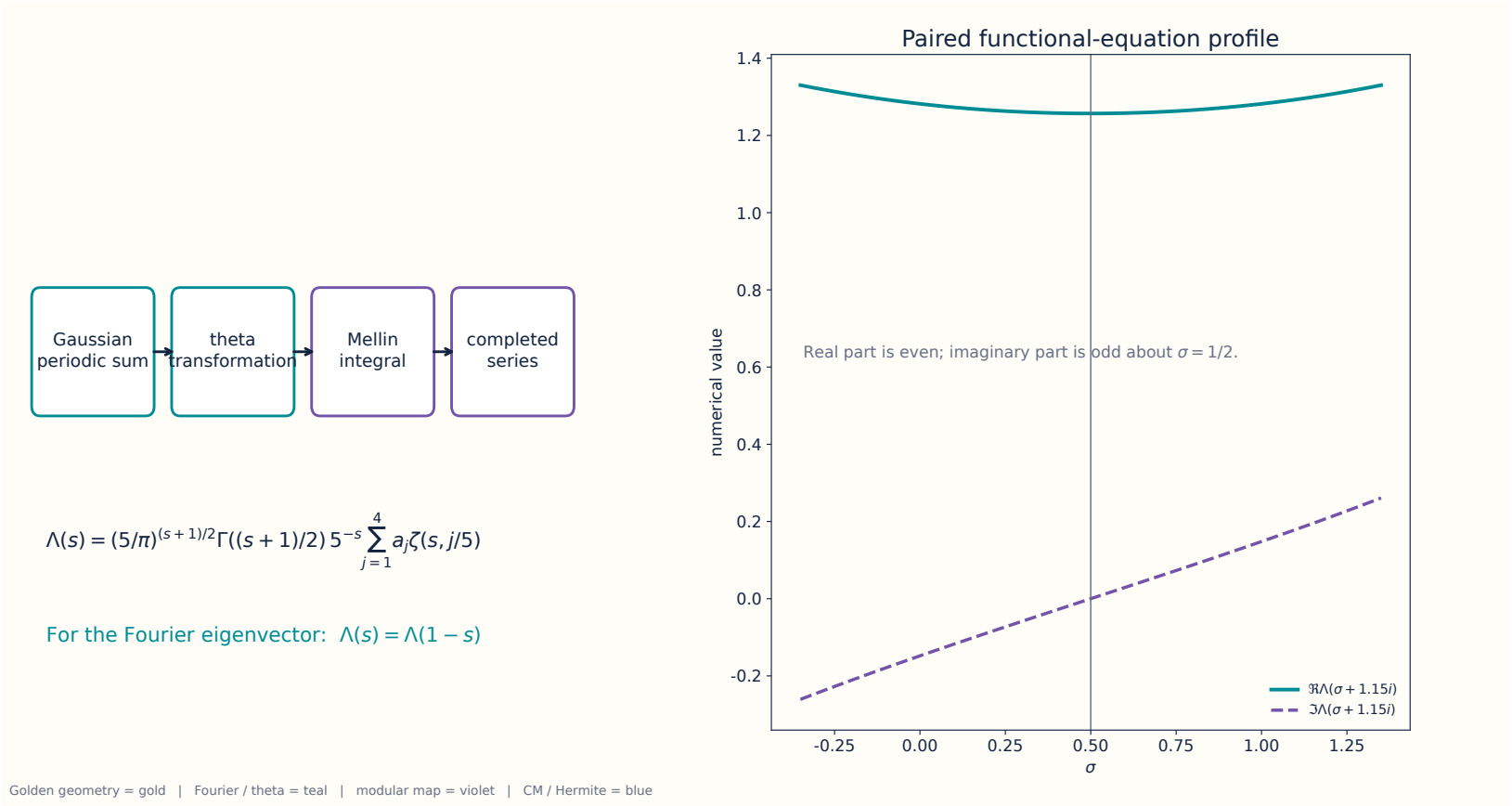


Figure 12: Mellin completion: the Gaussian sum, theta function, Mellin integral, and paired completed-series profile associated with Theorem 4.1.

Theorem 4.2. *For nonzero $a \in V_5$ and $\epsilon \in \{+1, -1\}$, $\Lambda_a(s) = \epsilon\Lambda_a(1-s)$ for all s iff $\mathcal{T}a = \epsilon a$. The positive line is $\mathbb{R}(1, \kappa, -\kappa, -1, 0)$, fixed by $a_1 = 1$; the negative line is $\mathbb{R}(1, -1/\kappa, 1/\kappa, -1, 0)$, and its completed value at $1/2$ is zero.*

Proof. Root-of-unity orthogonality gives $\mathcal{F}^2 a = a_{-n} = -a$, hence $\mathcal{T}^2 = I$. Solving the two eigenline equations of (5) gives the stated lines. One implication follows from Theorem 4.1. Conversely let $b = \mathcal{T}a - \epsilon a$. The assumed equation and that theorem make F_b vanish in a right half-plane. If m is its first nonzero coefficient, then for real $\sigma \to \infty$,

$$0 = m^\sigma F_b(\sigma) = b_m + \sum_{n>m} b_n (m/n)^\sigma.$$

The bounded tail tends to zero because

$$\left| \sum_{n>m} b_n (m/n)^\sigma \right| \le C m^\sigma \int_m^\infty x^{-\sigma} dx = \frac{Cm}{\sigma - 1}.$$

Thus $b_m = 0$, a contradiction. Central vanishing is immediate for $\epsilon = -1$. □

For a general vector, put $a_\pm = (a \pm \mathcal{T}a)/2$. Its completion is the sum of a symmetric and an antisymmetric part. If both components are nonzero, neither scalar-sign equation holds. For example, the coordinate vector $(1, 0)$ is an eigenvector of neither sign. The normalized $+1$ line is selected by $a_1 = 1$ and gives the Davenport–Heilbronn sequence.

Here $a_6 = 1$ but $a_2 a_3 = -\kappa^2$, so the normalized $+1$ sequence is nonmultiplicative. The off-critical-line zero result is classical and externally attributed [1]; this is only a classification inside V_5.

5 Conclusion

The surface calculation supplies the golden coordinate, the CM certificate recovers it from the actual τ_{163} value, and $A = JMJ$ supplies the analytic payoff at i. The recovered golden scalar determines a reflection, whose positive eigendirection determines the analytic symmetry. The explicit maps connect the geometric ratio, arithmetic field and functional equation through transformations that can each be checked. A precise remaining question is whether an N-dependent ruled-surface family can be constructed to carry the odd-N Fourier mechanism below.

A Exact certificate and square witness

Put $U = v - v^{-1}$ and

$$\alpha(U) = U^3 - a_1 U^2 + (3 - a_2)U - (2a_1 + a_3),$$
$$\beta(U) = -b_1 U^2 - b_2 U - (2b_1 + b_3).$$

The integer coefficients are

$$a_1 = 131268706320384342, \qquad b_1 = 58705150130167150,$$
$$a_2 = 2896632145572621515, \qquad b_2 = 1295413276662289800,$$
$$a_3 = 15717124384666951860, \qquad b_3 = 7028911706986972100.$$

The reconstruction polynomials are

$$A_{163}(X) = X^6 - a_1X^5 - a_2X^4 - a_3X^3 + a_2X^2 - a_1X - 1,$$
$$B_{163}(X) = -b_1X^5 - b_2X^4 - b_3X^3 + b_2X^2 - b_1X.$$

Exact expansion gives $P_{-640320^3} = A_{163}^2 - 5B_{163}^2$ and $\gcd(A_{163}, B_{163}) = 1$, equivalently $P_{-640320^3}(v) = v^6(\alpha(U)^2 - 5\beta(U)^2)$.

Define $G(q)$ as the positive product in (3), so $v = qG(q)^5$. For $q = -\mathrm{e}^{-\pi\sqrt{163}}$ every factor is positive and the phase is $r(\tau_{163}) = \mathrm{e}^{\pi\mathrm{i}/5}(-v)^{1/5}$. The rational estimates

$$3.14 < \pi < 22/7, \quad 12.76 < \sqrt{163} < 12.77, \quad \mathrm{e}^{40} > 23 \cdot 10^{16}, \quad \mathrm{e}^{41} < 7 \cdot 10^{17},$$

$$|\log G(q)| \le \frac{|q|}{(1-|q|)^2}$$

give $40 < \pi\sqrt{163} < 41$, $5|\log G(q)| < 1/1000$, and

$$-5 \cdot 10^{-18} < v < -10^{-18}.$$

Exact interval Horner arithmetic gives $A_{163}(v) < 0 < B_{163}(v)$; a Sturm count gives one root. The named checker `verification/baseline_cm.py` replays these finite facts and `CM_REPLAY_RESULTS.json` records separate 100/200-digit diagnostics; neither is an analytic proof.

With $S = \sqrt{5}$, define

$$\begin{aligned} w(U) = \frac{1}{1574580000}[&(312119004989 - 139583862445S)U^2 \\ &+ (6858643496508 - 3067324783740S)U \\ &+ 37679427598544 - 16850735555920S]. \end{aligned}$$

Exact reduction by $\alpha(U) + S\beta(U)$ and $S^2 - 5$ proves

$$U^2 + 4 = 163(1 + c_0)w(U)^2.$$

On the selected branch $U > 1$, the leading and linear coefficients are negative and $w(1) < 0$, hence $w'(U) < 0$ and $w(U) < 0$.

Corollary A.1. *For $\delta = \sqrt{163(1+c_0)} > 0$, $\mathbb{Q}(v) = \mathbb{Q}(U)(\delta)$.*

Proof. The coefficient formulas also give $A_{163}(v) = v^3\alpha(U)$ and $B_{163}(v) = v^3\beta(U)$. Hence $S = -\alpha(U)/\beta(U) \in \mathbb{Q}(U)$ and $w(U) \in \mathbb{Q}(U)$. Since $U \in \mathbb{Q}(v)$, both also belong to $\mathbb{Q}(v)$. Since $2v - U = v + v^{-1} < 0$ and $w(U) < 0$, $\delta = (2v - U)/w(U) \in \mathbb{Q}(v)$. Conversely $v = (U + \delta w(U))/2 \in \mathbb{Q}(U)(\delta)$. □

The exact remainder and the sign comparisons are replayed by `verification/baseline_square_certificate.py`. The selected radical has a direct pentagonal interpretation:

$$\delta = 2\sqrt{163}\sin(\pi/5), \qquad \phi\delta = 2\sqrt{163}\sin(2\pi/5).$$

Indeed $1 + c_0 = (5 - \sqrt{5})/2 = 4\sin^2(\pi/5)$ and $2\cos(\pi/5) = \phi$. The two lengths are the side and diagonal of a regular pentagon of circumradius $\sqrt{163}$; their squared ratio is c_0. Eliminating S from $\delta^2 = 163(5 - S)/2$ gives $\delta^4 - 815\delta^2 + 132845 = 0$, a different polynomial from the norm quartic for κ.

B Odd-N extension

For odd $N > 1$, let V_N be real odd N-periodic sequences and define

$$(\mathcal{F}_N a)_m = N^{-1/2} \sum_{n=1}^{N} a_n \mathrm{e}^{2\pi \mathrm{i} mn/N}, \qquad \mathcal{T}_N = -\mathrm{i}\mathcal{F}_N.$$

Its real sine matrix is

$$\frac{2}{\sqrt{N}} \big(\sin(2\pi mn/N)\big)_{1 \le m,n \le (N-1)/2}.$$

Orthogonality gives $\mathcal{F}_N^2 a = a_{-n} = -a$, so $\mathcal{T}_N^2 = I$; no unique positive line is claimed. With

$$\Theta_{a,N}(x) = \sum_{n \in \mathbb{Z}} n a_n \mathrm{e}^{-\pi n^2 x/N},$$

$$\Lambda_{a,N}(s) = \frac{1}{2} \int_0^\infty \Theta_{a,N}(x) x^{(s+1)/2} \frac{dx}{x},$$

the preceding residue-class proof, replacing 5 by N, gives

$$\Theta_{a,N}(x) = x^{-3/2} \Theta_{\mathcal{T}_N a,N}(1/x),$$

$$\Lambda_{a,N}(s) = \left(\frac{N}{\pi}\right)^{(s+1)/2} \Gamma((s+1)/2) F_a(s) = \Lambda_{\mathcal{T}_N a,N}(1-s).$$

The series expression initially holds for $\Re s > 1$. The integral is entire; the same endpoint bounds and local-uniform differentiation argument apply. This is a finite-Fourier theorem, distinct from an unconstructed N-dependent surface.

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